

CM0859 – MT5009 Type Theory Constructivism

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Preliminaries

- “Textbook”

“Lecture Notes on What is (Constructive) Logic?” (Pfenning 2023).

- Other references

- *Introduction to Metamathematics* (Kleene [1952] 1974).

- *Constructivism in Mathematics. An Introduction. Volume I* (Troelstra and van Dalen 1988).

The Crisis in the Foundations of Mathematics

Paradoxes $\Rightarrow$ Crisis $\Rightarrow$ $\left\{ \begin{array}{ll} \text{Logicism} & (\text{Russell and Whitehead}) \\ \text{Formalism} & (\text{Hilbert}) \\ \text{Intuitionism} & (\text{Brouwer}) \end{array} \right.$

The Crisis in the Foundations of Mathematics

Logicism (Russell and Whitehead)

“The logicistic thesis is that mathematics is a branch of logic. The mathematical notions are to be defined in terms of the logical notions. The theorems of mathematics are to be proved as theorems of logic.” (Kleene [1952] 1974, p. 43)

The Crisis in the Foundations of Mathematics

Formalism (Hilbert)

“Classical mathematics shall be formulated as a formal axiomatic theory, and this theory shall be proved to be consistent, i.e. free from contradiction.” (Kleene [1952] 1974, p. 53)

The Crisis in the Foundations of Mathematics

Intuitionism (Brouwer)

“Intuitionism is based on the idea that mathematics is a creation of the mind. The truth of a mathematical statement can only be conceived via a mental construction that proves it to be true.” (Iemhoff 2024)

The Crisis in the Foundations of Mathematics

Conceptions of the infinite

(i) Non-Intuitionism

“The infinite is treated as **actual** or **completed** or **extended** or **existential**. An infinite set is regarded as existing as a completed totality, prior to or independently of any human process of generation or construction, and as though it could be spread out completely for our inspection.” (Kleene [1952] 1974, p. 48)

(ii) Intuitionism

“The infinite is treated only as **potential** or **becoming** or **constructive**. The recognition of this distinction, in the case of infinite magnitudes, goes back to Gauss, who in 1831 wrote, ‘I protest . . . against the use of an infinite magnitude as something completed, which is never permissible in mathematics.’ (Werke VIII p. 216.)” (Kleene [1952] 1974, p. 48)

Constructivism

Some differences with classical logic

- (i) Rejection of the principle of exclude middle (*tertium non datur*).

$$\vdash A \vee \neg A, \text{ for all formula } A.$$

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(ii) A proof of an existential formula $\exists x.A(x)$ must include a **witness** t such as $A(t)$ is true.

(continued on next slide)

Constructivism

Some differences with classical logic

(iii) Rejection of proofs by contradiction

Proof by contradiction
(or *reductio ad absurdum*)

$$[\neg A]$$
$$\frac{\vdots}{\perp}$$
$$\frac{}{A}$$

Proof of negation (Bauer 2017)

$$[A]$$
$$\frac{\vdots}{\perp}$$
$$\frac{}{\neg A}$$

Non-Constructive Proofs

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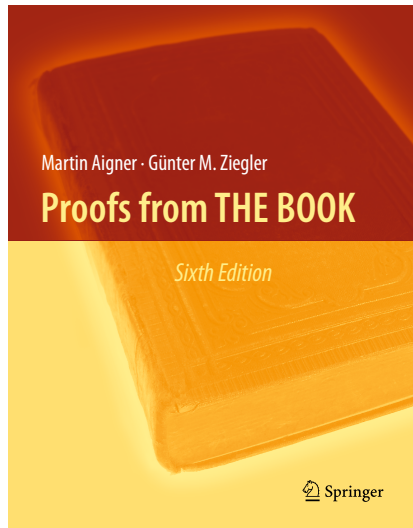
Proof (using the principle of exclude middle)

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Question

Could you give me two irrational numbers r, s such that r^s is rational?

Non-Constructive Proofs



Non-Constructive Proofs

Example

There are an infinity number of primes.

Non-Constructive Proofs

Example

There are an infinity number of primes.

Proof (by contradiction)

“**Euclid’s proof.** For any finite set $\{p_1, \dots, p_r\}$ of primes, consider the number $n = p_1 p_2 \cdots p_r + 1$. This n has a prime divisor p . But p is not one of the p_i : otherwise p would be a divisor of n and of the product $p_1 p_2 \cdots p_r$, and thus also of the difference $n - p_1 p_2 \cdots p_r = 1$, which is impossible. So a finite set $\{p_1, \dots, p_r\}$ cannot be the collection of **all** prime numbers.” (Aigner and Ziegler [1998] 2018, p. 3)

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Question

Could you give me an infinite list of primes?

Non-Constructive Proofs

Definition

The **Cartesian product** (or **generalised product**) of a family of sets $\langle A_i \mid i \in I \rangle$ is defined by

$$\prod_{i \in I} A_i := \left\{ f \mid f : I \rightarrow \bigcup_{i \in I} A_i \text{ and } \forall i (i \in I \rightarrow f(i) \in A_i) \right\}.$$

Non-Constructive Proofs

The axiom of choice

The axiom of choice is other source of non-constructive proofs: Let $\langle H_i \mid i \in I \rangle$ be a family a sets. If $H(i) \neq \emptyset$ for all $i \in I$, then $\times_{i \in I} H(i) \neq \emptyset$.

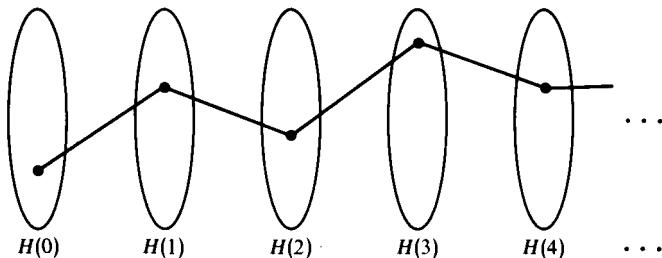


Illustration of the axiom of choice.[†]

[†]Figure source: (Enderton 1977, Fig. 11).

The Brouwer-Heyting-Kolmogorov (BHK) Interpretation

Logical constants

$\wedge$	(and)	conjunction
$\vee$	(or)	(inclusive) disjunction
$\supset$	(if __, then __)	conditional
$\perp$	(falsity)	bottom, falsum
$\forall x$	(for every x)	universal quantifier
$\exists x$	(there exists a x)	existential quantifier

Definition

We define negation by $\neg A := A \supset \perp$.

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- There is no proof of $\perp$.
- A proof of $\neg A$ is a function (method, program) which transforms a (hypothetical) proof of A into a contradiction.
- A proof of $\exists x.A$ is a construction of a witness d and a proof of $A(d)$.
- A proof of $\forall x.A$ is a function (method, program) which takes an arbitrary individual d into a proof of $A(d)$.

Connection Between Proofs and Programs

Example

We define the follow predicates on natural numbers:

$$\text{even}(x) := \exists y. x = 2y,$$

$$\text{odd}(x) := \exists y. x = 2y + 1.$$

Prove that $\forall x. \text{even}(x) \vee \text{odd}(x)$.

Proof (by induction on x)

(Whiteboard)

Connection Between Proofs and Programs

Haskell function from the proof that $\forall x.\text{even}(x) \vee \text{odd}(x)$

```
1  data Nat = Zero | Succ Nat
2
3  data E0 = Even | Odd
4    deriving Show
5
6  isEvenOrOdd :: Nat -> E0
7  isEvenOrOdd Zero      = Even
8  isEvenOrOdd x@(Succ x') = case isEvenOrOdd x' of
9    Even -> Odd
10   Odd  -> Even
```

References

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-  Rosalie Iemhoff (2024). Intuitionism in the Philosophy of Mathematics. In: The Stanford Encyclopedia of Philosophy. Ed. by Edward N. Zalta and Uri Nodelman. Summer 2024. Metaphysics Research Lab, Stanford University. URL: <https://plato.stanford.edu/archives/sum2024/entries/intuitionism> (cit. on p. 6).
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References



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A. S. Troelstra and D. van Dalen (1988). *Constructivism in Mathematics. An Introduction*. Volume I. Vol. 121. *Studies in Logic and the Foundations of Mathematics*. North-Holland (cit. on p. 2).